

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0782

Gardner 230 kV
May 21, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0782. The Interconnection Customer proposes a 1169.8 MW natural gas combined cycle facility interconnecting at the Gardner 230 kV substation in the Illinois area. The request was received on April 25, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 231 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$104,604,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gardner / Top Crop 230 kV L	Grand Ige Battery Projects / Gardner 230 kV	291.4	454	99.9
Grand Ridge Battery Projects / Gardner 230 kV	Top Crop / Gardner 230 kV	260.4	508	99.9
Top Crop 500-230 kV Transformer Bank	Grand Ridge Battery Projects / Gardner 230 kV	166.4	705	95.0
Minnehaha / Napoleon Peaking Station 230 kV L	Troy Energy Llc Substation / Richland 500 kV	125.4	678	13.0
Darlington Rd / Wayne Trace 100 kV Line	Richland / Dekalb 500 kV L	125.1	258	6.0
Muskogee / Darlington Rd 100 kV Line	Richland / Dekalb 500 kV	122.9	231	6.7
Sta S Van Nuys / Minnehaha 230 kV	Richland / Troy Energy Llc 500 kV	119.5	786	16.9
Swan Vly / Eastown Rd 230 kV line	Swan Valley / Trego 500 kV Line	115.9	614	6.3
Truman / Waldo 230 kV	Trego / Beaver 500 kV	115.5	546	9.9
Joso / Onancock Diesel Generation Facility 230 kV	Eufaula / Beaver 500 kV	114.9	1,316	14.3
New Departure Substation / Mason St 230 kV	Beaver / Trego Substation 500 kV	114.6	842	7.8
Onancock Diesel Generation Facility / Beaver 230 kV	Eufaula / Beaver 500 kV	114.4	1,315	10.8
Mason St 500/230kV XFMR	Trego / Beaver 500 kV L	111.0	1,308	11.9
Adams / Pickford 230 kV	Beaver / Trego 500 kV	109.8	855	8.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Adams Substation / Onancock Diesel Generation Facility 230 kV L	Trego / Beaver 500 kV L	109.2	1,686	15.9
Post Rock / Myrtle Point 230 kV	Trego / Beaver Substation 500 kV	109.0	1,406	12.8
Pickford / Herad-Twin Cities 230 kV	Trego / Beaver 500 kV	108.7	1,138	9.7
Waldo Substation 500 kV/230 kV XFMR	Trego / Beaver 500 kV	108.7	716	9.8
Highgate / Herald-Twin Cities 230 kV	Mason St 500-230 kV Auto	108.6	857	7.0
Adams / New Departure 230 kV	Trego / Beaver 500 kV line	108.5	864	7.8
Herald-Twin Cities / Mason St 230 kV	New Departure / Mason St 230 kV L	108.3	610	5.5
Post Rock / Newport Tie 230 kV	Trego / Beaver 500 kV Line	108.3	1,358	13.2
Station S Van Nuys / Gould 230 kV	Troy Energy Llc / Richland 500 kV	107.6	511	6.6
Sidney A Murray Jr Hydroelectric / 24Th & Gardendale 230 kV	Richland Substation / Dekalb 500 kV line	107.0	492	9.1
Kendallville / Farmersville 100 kV	Dekalb / Jones Chapel 500 kV	106.9	205	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gardner / Top Crop 230 kV L	Rebuild line with higher-capacity conductor	\$9,689,000	93% (\$9,015,000)
Grand Ridge Battery Projects / Gardner 230 kV line	Rebuild line with higher-capacity conductor	\$7,958,000	93% (\$7,404,000)
Top Crop 500/230 kV Auto	Replace transformer with higher-rated unit	\$21,046,000	93% (\$19,581,000)
Minnehaha / Napoleon Peaking Sta 230 kV	Rebuild line with higher-capacity conductor	\$49,629,000	17% (\$8,313,000)
Darlington Rd / Wayne Trace 100 kV Line	Rebuild line with higher-capacity conductor	\$2,342,000	17% (\$393,000)
Muskogee / Darlington Road 100 kV	Rebuild line with higher-capacity conductor	\$19,531,000	17% (\$3,273,000)
Sta S Van Nuys / Minnehaha 230 kV	Rebuild line with higher-capacity conductor	\$15,805,000	17% (\$2,646,000)
Swan Vly / Eastown Rd 230 kV line	Rebuild line with higher-capacity conductor	\$13,441,000	17% (\$2,259,000)
Truman / Waldo 230 kV	Rebuild line with higher-capacity conductor	\$25,035,000	17% (\$4,208,000)
Onancock Diesel Generation Facility / Joso Substation 230 kV	Rebuild line with higher-capacity conductor	\$23,259,000	17% (\$3,900,000)
New Departure Substation / Mason St 230 kV	Rebuild line with higher-capacity conductor	\$53,340,000	17% (\$8,948,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Onancock Diesel Generation Facility / Beaver 230 kV	Rebuild line with higher-capacity conductor	\$19,415,000	17% (\$3,253,000)
Mason St 500/230kV XFMR	Replace transformer with higher-rated unit	\$30,237,000	17% (\$5,075,000)
Adams / Pickford 230 kV	Reconductor line	\$15,374,000	17% (\$2,577,000)
Adams Substation / Onancock Diesel Generation Facility 230 kV L	Reconductor line	\$19,904,000	17% (\$3,338,000)
Post Rock / Myrtle Point 230 kV	Reconductor line	\$11,104,000	17% (\$1,860,000)
Pickford / Herad-Twin Cities 230 kV	Reconductor line	\$20,148,000	17% (\$3,379,000)
Waldo Substation 500 kV/230 kV XFMR	Replace transformer with higher-rated unit	\$20,779,000	17% (\$3,491,000)
Highgate / Herald-Twin Cities 230 kV	Reconductor line	\$15,792,000	17% (\$2,650,000)
Adams / New Departure 230 kV	Reconductor line	\$11,129,000	17% (\$1,867,000)
Herald-Twin Cities / Mason St 230 kV	Reconductor line	\$12,142,000	17% (\$2,040,000)
Post Rock / Newport Tie 230 kV	Reconductor line	\$11,091,000	17% (\$1,858,000)
Station S Van Nuys / Gould 230 kV	Reconductor line	\$7,427,000	17% (\$1,247,000)
Sidney A Murray Jr Hydroelectric / 24Th & Gardendale 230 kV	Reconductor line	\$10,032,000	17% (\$1,679,000)
Kendallville / Farmersville 100 kV	Reconductor line	\$2,089,000	17% (\$350,000)
	Total allocated to project		\$104,604,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.